Before we see how Bayesian optimization solves this optimization problem, let's look at two baseline strategies. The first is a random search, where we uniformly sample between -5 and 5, and whatever points we end up with are the locations that we will evaluate the objective at. Figure 1.7 is the result of one such scheme. The point with the highest value found here is at roughly 4, having the value of 3.38.

SIDEBAR **How random search works**

Random search involves choosing points uniformly at random within the domain of our objective function. That is, the probability that we end up at a point within the domain is equal to the probability that we end up at any other point.

One thing you might find unsatisfactory about these randomly sampled points is that many of them (by chance) fall into the region around 0. Intuitively, it is more beneficial to "spread out" our evaluations, so that we learn more about the objective function.

This idea of "spreading out" evaluations leads us to the second baseline: grid search. Here, we divide our search space into evenly spaced segments and evaluate at the endpoints of those segments like in Figure 1.8.

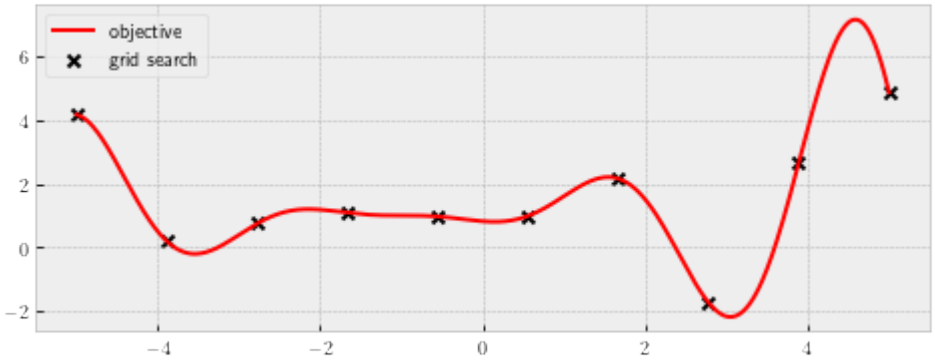


Figure 1.8 Grid search is still inefficient at narrowing down a good region.

The best point from this search is the very last point on the right at 5, evaluating at roughly 4.86. Better than random search but still missing the actual global optimum.

Now, we are ready to look at Bayesian optimization in action! Bayesian optimization starts off with a randomly sampled point, just like random search, shown in Figure 1.9.

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